4 **(a)** Fig.4.1 shows a string stretched between two fixed points P and Q.

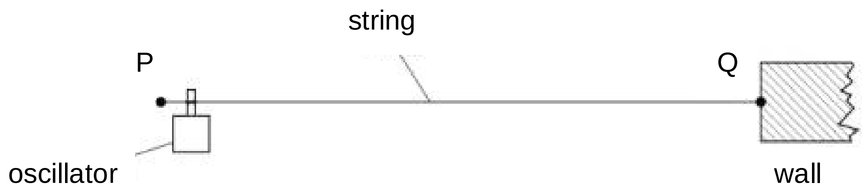

Fig. 4.1

An oscillator is attached near end P of the string. End Q is fixed to a wall. The oscillator has a frequency of 50.0 Hz.

The stationary wave produced on PQ at an instant time *t* is shown in Fig. 4.2. Each point on the string is at its maximum displacement.

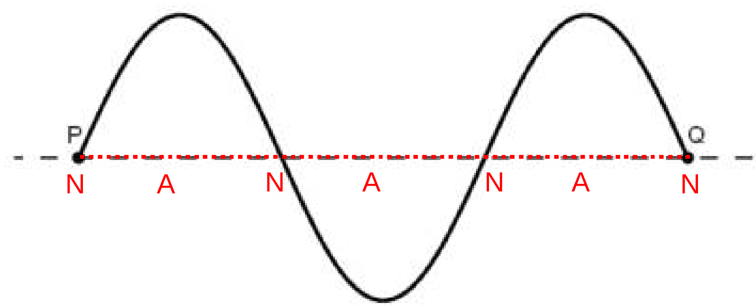

Fig. 4.2

(i) On Fig. 4.2, label all the nodes with the letter **N** and the antinodes with the letter **A** along the dotted line PQ. [1]

(ii) On Fig 4.2, draw the stationary wave at (t + 5.0 ms). [1]

Since T = 1/f = 1/50 = 20 ms,
5 ms is ¼ of a period